Sec: SR.IIT_*COSC(MODEL-A&APEX) | GTM-1 | Date: 03-08-24
Time: 3 Hrs | JEE-MAIN | Max. Marks: 300

KEY SHEET

PHYSICS

1	**B**	2	**A**	3	**A**	4	**B**	5	**C**
6	**C**	7	**C**	8	**B**	9	**D**	10	**B**
11	**D**	12	**A**	13	**B**	14	**C**	15	**D**
16	**A**	17	**D**	18	**A**	19	**A**	20	**C**
21	**225**	22	**60**	23	**2**	24	**6**	25	**9**

CHEMISTRY

26	**C**	27	**A**	28	**C**	29	**A**	30	**B**
31	**D**	32	**A**	33	**C**	34	**D**	35	**C**
36	**A**	37	**C**	38	**B**	39	**A**	40	**D**
41	**B**	42	**A**	43	**B**	44	**B**	45	**C**
46	**5**	47	**34**	48	**314**	49	**11**	50	**96**

MATHEMATICS

51	**D**	52	**B**	53	**A**	54	**A**	55	**D**
56	**A**	57	**B**	58	**A**	59	**C**	60	**D**
61	**C**	62	**A**	63	**B**	64	**D**	65	**A**
66	**C**	67	**C**	68	**B**	69	**A**	70	**B**
71	**9**	72	**3**	73	**5**	74	**19**	75	**49**

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SOLUTIONS

PHYSICS

1. $$I=\frac{m}{4}\times\frac{a^2}{3}\times 2+\frac{m}{4}\times\left(\frac{a^2}{12}+\left(\frac{a}{2}\right)^2+a^2\right)\times 2$$

$$=\frac{ma^2}{6}+\frac{m}{2}\left(\frac{a^2+3a^2+12a^2}{12}\right)$$

$$=\frac{ma^2}{6}+\frac{2}{3}ma^2=\frac{ma^2+4ma^2}{6}=\frac{5ma^2}{6}$$

2. $$\frac{dH}{dt}=-2\pi x^2k\frac{dT}{dx}\quad\Rightarrow\frac{dH}{dt}\times\int_{2.5}^{3}\frac{dx}{x^2}=2\pi k\int_{-40}^{5}dT$$

$$\frac{dH}{dt}=\frac{2\pi\times k\times 45}{\frac{1}{2.5}-\frac{1}{3}}=2.25\pi kW$$

3. $$\lambda=\frac{v}{\sqrt{2}\pi d^2nN_a}=\frac{RT}{\sqrt{2}\pi d^2pN_a}\quad=\frac{\frac{25}{3}\times 300}{\sqrt{2}\times\pi\times 10^{-20}\times 10^5\times 6\times 10^{23}}$$

$$=\frac{25}{6\sqrt{2\pi}}\times 10^{-6}m$$

$$x^3=\frac{v}{nN_a}$$

$$\Rightarrow x=\left(\frac{RT}{PN_a}\right)^{1/3}$$

$$=\left(\frac{\frac{25}{3}\times 300}{10^5\times 6\times 10^{23}}\right)^{1/3}$$

4. $$T=2\pi\sqrt{\frac{\ell}{g}}$$

In summer, length will increase due to expansion, hence for same time period we have to decrease ℓ in summer.

5. $$v=108\times\frac{5}{18}=30\ m/s$$

$$f_1=\frac{C}{C-v}f_0$$

$$550=\frac{330}{330-30}f_0$$

$$f_0=500\ Hz$$

$$f=\frac{330}{330+30}\times 500=\frac{33}{36}\times 500=\frac{5500}{12}$$

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6. (1) $2.3056 \rightarrow 4$ decimals
$10.138 \rightarrow 3$ decimals
$-7.4671 \rightarrow 4$ decimals
4.9765
Answer should have 3 decimals

(2) $2.38 \times 1.0 = 2.38 \rightarrow$ answer should have 2 significant digits

(3) $\frac{8.05}{3.1} = 2.59 \approx 2.6$ answer should have 2SD

(4) $1.11 - 0.1 = 1.01 \approx 1.0$ but it is in an intermediate step so we keep 1 digit extra
$1.01 \times 9.0 = 9.09 \approx 9.1 \rightarrow$ both 1.01 and 9.0 have 2SD

7. $a = \frac{v_0^2}{R}$

After 10 kicks, $v = v_0 + \frac{10v_0}{100} = 1.1v_0$

$a = \frac{(1.1v_0)^2}{R_1} = \frac{v_0^2}{R} \Rightarrow R_1 = 1.21\,R$

8. $\tan\theta = \frac{v_y}{v_x} = \frac{u_y - gt}{u_x}$

$u_x = \frac{4}{1}\text{m/s}$

$3 = u_y \times 1 - \frac{1}{2} \times 10 \times 1^2$

$u_y = 8\text{ m/s}$

$\Rightarrow \tan\theta = \left|\frac{8-10}{4}\right| = \left|-\frac{1}{2}\right|$

9. $B = mg \Rightarrow B_1 = B_2$
$\rho_1 V_1 g = B = \rho_w v_{LD} g$

$\frac{\rho_1}{\rho_w} = \frac{V_{LD}}{V_1} = f_1$

$\frac{\rho_2}{\rho_w} = f_2 \Rightarrow \rho_2 > \rho_1$

10. $i = \frac{v}{R+G}$

$\frac{i}{2} = \frac{v}{\frac{SG}{S+G}} \times \frac{S}{S+G} \Rightarrow \frac{2S}{SG+RG+RS} = \frac{1}{R+G}$

$SG + SR = RG$

$\frac{SR}{R-S} = G$

$R >> S$

$\Rightarrow G \approx S$

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11. $v = \frac{2}{9}\frac{r^2 g(\rho-\sigma)}{\eta}$

$v' = \frac{2}{9} r^2 g\left(\frac{\rho - 2\sigma}{\eta}\right)$

12. $WD = \Delta QV \Rightarrow \Delta Q = 4\mu C$

$Q_i = \frac{2\times 4}{2+4}\times 6 = 8\mu C,\ Q_f = 12\mu C = C_{eq}\times 6$

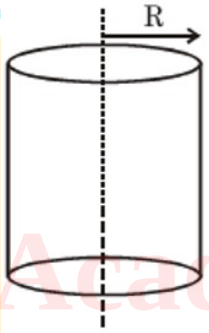
$C_{eq} = 2\mu F \Rightarrow C_1 = C_2 = 4\mu F$

$k = \frac{C_1}{C_0} = 2$

13. $\phi = \frac{q}{24\varepsilon_0} - \frac{q}{\varepsilon_0 4\pi}\frac{2\pi\left(1-\frac{1}{\sqrt{2}}\right)}{4}$

$= \frac{q}{24\varepsilon_0} - \frac{q}{8\varepsilon_0} + \frac{q}{8\sqrt{2}\varepsilon_0} = \frac{q}{8\sqrt{2}\varepsilon_0} - \frac{q}{12\varepsilon_0} \approx 5\ Nm^2/C$

14. Applying Ampere's Law



For inside

$B2\pi r = \mu_0 J\pi r^2$

$\Rightarrow B \propto r$

For outside

$B2\pi r = \mu_0 J\pi R^2$

$\Rightarrow B \propto \frac{1}{r}$

15. Heat will develop due to eddy current which will oppose the motion and hence a force is required.

16. $Q_{(i)} = (M_{Br} - M_{Kr})\ c^2 > 0$

$Q_{(ii)} = (M_{Br} - M_{Sc} - 2M_e)\ c^2 > 0$

$Q_{(ii)} = (M_{Br} - M_{Se})\ c^2 > 0$

17. In an inelastic collision kinetic energy is lost. There is no physical meaning of heat and sound at atomic level.

18.

$P = i^2R = \frac{v_0^2}{2(x_0^2 + (R_1+R_2)^2)} \times R_2$

$= \frac{900\times 8}{2\left(\frac{1}{5000\times 2\pi}\times 10^6\right)^2 + 20^2}$

$= \frac{3600}{1000+400} = \frac{3600}{1400} = 2.57W$

20. Pressure of reflection $= \frac{2I}{C} = \frac{2Power}{AC}$

$= \frac{2\times 1.5\times 10^9}{1.3\times 10^{-6}\times 3\times 10^8} = \frac{2\times 1.5}{1.3\times 3}\times 10^7 = \frac{1}{1.3}\times 10^7$

$= 7.7 \times 10^6$

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21. $v = \frac{\omega}{k} = \frac{5\pi \times 30}{\pi} = 150 = \sqrt{\frac{T}{\frac{1}{100}}} \quad \Rightarrow T = \frac{2250000}{100} = 225\,N$

22.

Initially $V_R = V_{C_1} = 2.5 \times 4 = 10V$

$Q_1 = CV = 3 \times 10 = 30\ \mu C$

$U_1 = \frac{1}{2} \times 3 \times 10^2 = 150\ \mu J$

Finally $V_{C_1} = V_{C_2} = \frac{Q_1}{C_1 + C_2} = \frac{30}{5} = 6\ V$

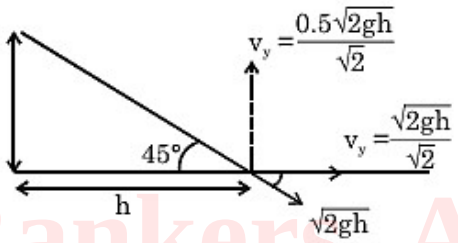
$U_2 = \frac{1}{2} \times (C_1 + C_2) \times V^2$

$= \frac{1}{2} \times 5 \times 36 = 90\ \mu J$

$\therefore$ Heat liberated = 150 – 90 = 60 μJ

23.

Sol.



$R = u_x \times T$

$= \sqrt{gh} \times \frac{0.5\sqrt{gh} \times 2}{g} = h$

$\Rightarrow AB = h + h = 2h$

24.

. Magnitude of viscous force, $F = \eta A = \frac{dv}{dr}$

$\Rightarrow$ viscous force per unit area $\sigma = \frac{F}{A} = \eta \frac{dv}{dr}$

$v = v_0\left(1 - \frac{r^2}{R^2}\right)$

$\Rightarrow \frac{dv}{dr} = -\frac{2V_0 r}{R^2}$

$\Rightarrow \sigma = \eta \frac{2v_0 r}{R^2}$ ..... (i)

Volume rate of flow, Q

consider an annular element at r from axis, width dr.

$dA = 2\pi r dr;\ dQ = v.dA = v_0\left(1 - \frac{r^2}{R^2}\right)2\pi r dr$

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$$Q = \int dQ = 2\pi v_0 \left[\frac{r^2}{2} - \frac{r^4}{4R^2}\right]_0^R = \frac{\pi}{2} R^2 v_0$$

$$\Rightarrow \quad v_0 = \frac{2Q}{\pi R^2}$$

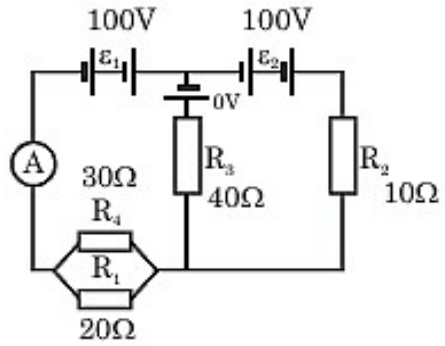
$$\therefore \quad \text{(i)} \Rightarrow \sigma = \eta \frac{4Q}{\pi R^4} r, \quad R = 0.1 \text{ m}$$

At $r = 0.04$ m,

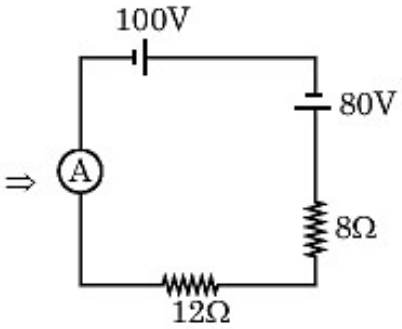
$$\sigma = (0.75)4 \times \frac{\pi}{2} \times 10^{-2} \times \frac{0.04}{\pi \times 10^{-4}}$$

$$= 6 \text{ Nm}^{-2}$$

25.



$$\varepsilon_{eq} = \frac{\frac{0}{40} + \frac{100}{10}}{\frac{1}{40} + \frac{1}{10}} = \frac{10}{\frac{5}{40}} = 80\text{V}$$



$$R_{eq} = \frac{400}{50} = 8\Omega$$

$$R'_{eq} = \frac{30 \times 20}{30 + 20} = \frac{600}{50} = 12$$

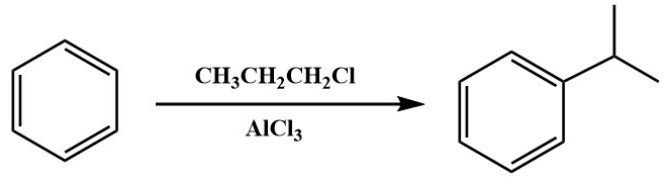
$$i = \frac{180}{12 + 8} = 9\text{A}$$

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CHEMISTRY

26. Perkins Reaction followed by esterification.

27. Conceptual

28. 

Major product due to carbocation rearrangement.

29. (1) $(CH_3)_3CCl$, $AlCl_3$; (2) HNO_3, H_2SO_4; (3) Br_2, $AlBr_3$; (4) Cl_2, $AlCl_3$
This sequence will give the desire product.

30. Of an ester and an oxo compound of analogous structure, the oxo compound is more enolized (stronger CH acid) than the ester, resulting in the oxo compound reacting with its negatively charged α-carbon atom and the ester with its positively polarized carbonyl carbon atom.

31. The isoelectric point of amino acids can fall into the acidic, neutral or basic pH range, depending on the number and pK value of the acidic and basic groups in the molecule. In solution, the acidic group transfers its proton to the basic group and a zwitterionic form is formed.

32. Preparation of Phenol and Acylation

33. Hint: $\frac{1}{\lambda} = RZ^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

34. Conceptual

35. $[F^{\ominus}]_0 = [F^{\ominus}]_{eq} + [HF]$

$$[Li^+] = \frac{Ksp}{[Li^+]} + \frac{Kw}{Ka}\frac{[F^{\ominus}]}{[OH^{\ominus}]}$$

$$[Li^+] = Ksp\left[1 + \frac{[H^{\ominus}]}{Ka}\right]$$

$$[Li^+] = 2\times10^{-12}\left[1 + \frac{10^{-4}}{10^{-4}}\right] = 4\times10^{-12}\ M$$

36. $A(g) + 2B(g) \rightarrow C(g) + D(g)$
Rate $= k[A][B]^2 = k(0.60(0.80)^2$ ……(i)
When $p_c = 0.20$ atm p_A is reduced to 0.40 and $p_B = 0.40$ (See stoichiometric representation)
Rate $= k[0.40]\ [0.40]^2$ ……(ii)
∴ (ii) divide by (i)

$$= \frac{0.40\times0.40\times0.40}{0.60\times0.80\times0.80} = \frac{1}{6}$$

37. $p_A = p_A^0 \times x_A = \text{Total pressure} \times y_A$

$p_B = p_B^0 \times x_B = \text{Total pressure} \times y_B$

Where x and y represent mole fraction in liquid and vapour phase respectively

$$\therefore \frac{p_B^0 x_B}{p_B^0 x_A} = \frac{y_B}{y_A},$$

$$\frac{p_B^0(1-x_A)}{p_A^0 x_A} = \frac{1-y_A}{y_A}$$

$$\frac{1}{x_A} - 1 = \frac{p_A^0}{p_B^0}\left(\frac{1}{y_A} - 1\right) = \frac{p_A^0}{p_B^0}\frac{1}{y_A} - \frac{p_A^0}{p_B^0}$$

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$$\text{or, } \frac{1}{x_A} = \frac{p_A^0}{p_B^0}\cdot\frac{1}{y_A} + \left(1 - \frac{p_A^0}{p_B^0}\right) = \frac{p_A^0}{p_B^0}\cdot\frac{1}{y_A} + \frac{p_B^0 - p_A^0}{p_B^0}$$

This is equation of straight line

$$\text{Slope} = \frac{p_A^0}{p_B^0}, \quad \text{Intercept} = \frac{p_B^0 - p_A^0}{p_B^0}$$

38. $\underset{1-\alpha}{PCl_5} \rightleftharpoons \underset{\alpha}{PCl_3} + \underset{\alpha}{Cl_2}$

$$\therefore K_p = \frac{\frac{\alpha}{1+\alpha}P \times \frac{\alpha}{1+\alpha}P}{\frac{1-\alpha}{1+\alpha}P} = \frac{\alpha^2 P}{1-\alpha^2}$$

$$\text{or, } K_p = \alpha^2 p \qquad \therefore \alpha = \sqrt{\frac{K_p}{P}} \quad \text{when } 1 - \alpha^2 = 1$$

39. $\Lambda^0_{M_{(HA)}} = \Lambda^0_{M_{(HCl)}} + \Lambda^0_{M_{(NaA)}} - \Lambda^0_{M_{(NaCl)}} = 450\ Scm^2\ mol^{-1}$

$$\underset{C-C\alpha}{HA} \rightleftharpoons \underset{C\alpha}{H^+} + \underset{C\alpha}{A^-}$$

Since $pH = 5 \Rightarrow [H^+] = 10^{-5}$

$C\alpha = 10^{-5}$

$$\alpha = \frac{\Lambda_m}{\Lambda'_m} = \frac{250}{450} = 0.56$$

$$K_a = \frac{C\alpha^2}{1-\alpha} = \frac{(C\alpha)(\alpha)}{(1-\alpha)} = \frac{10^{-5}(0.56)}{1-0.56}$$

$$K_a = \frac{0.56\times10^{-5}}{0.44} = 1.27\times10^{-5}$$

40. Conceptual
41. Conceptual
42. Conceptual
43. $2Na_2S_2O_3 + I_2 \longrightarrow \underset{(A)}{2NaI} + \underset{(B)}{Na_2S_4O_6}$, $NaI + KMnO_4(aq) \xrightarrow{\text{mild alkaline}} \underset{\text{(soluble) (C)}}{NaIO_3} + \underset{\text{Black brown ppt (D)}}{MnO_2}\downarrow$

$$MnO_2 + 4HCl \longrightarrow MnCl_2 + \underset{(E)}{Cl_2(gas)} + H_2O$$

44. Salt AB can be $AgNO_2$ or AgBr

$$Ag^+(aq) + H_2S \xrightarrow{HCl} \underset{\text{Black ppt}}{Ag_2S}$$

$$AgNO_2(s) + NH_4OH \longrightarrow \left[Ag(NH_3)_2\right]NO_2$$

$$AgBr(s) + NH_4OH \longrightarrow \left[Ag(NH_3)_2\right]Br$$

$$NO_2^- + \underset{\text{conc.}}{H_2SO_4} \longrightarrow HNO_2 \longrightarrow HNO_3 + NO \xrightarrow{O_2} \underset{\text{Brown gas}}{NO_2}$$

$$Br^- + \underset{\text{conc.}}{H_2SO_4} \longrightarrow Br_2 \uparrow_{\text{Brown gas}}$$

$$AgNO_2 / AgBr(s) + Na_2S_2O_3(aq) \longrightarrow Na_3\left[Ag(S_2O_3)_2\right] + NaNO_2 / NaBr$$

$$CuBr_2 + Conc.H_2SO_4 \rightarrow Br_2 \uparrow_{Brown\ gas}$$

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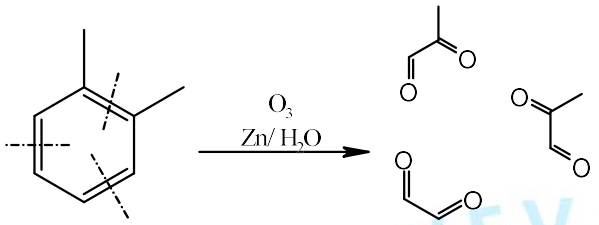
$CuBr_2 + NH_4OH \rightarrow \underset{Soluble\ complex}{\left[Cu\left(NH_3\right)_4\right]^{2+}}$

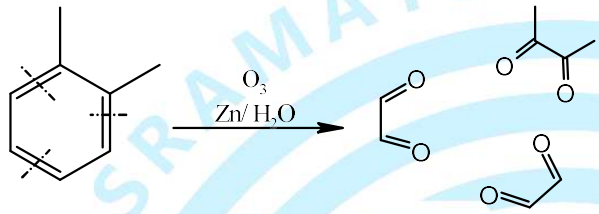
$AgNO_2\ or\ AgBr + Na_2S_2O_3 \rightarrow \left[Ag\left(S_2O_3\right)_2\right]^{3-}$

$CuBr_2 + Na_2S_2O_3 \rightarrow \left[Cu_6\left(S_2O_3\right)_5\right]^{4-} \rightarrow Soluble$

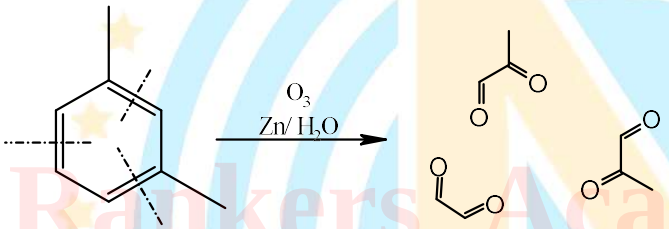
45. Conceptual

46.





3 different product
X = 3



2 different product
Y = 2

47. Products are methyl alcohol and another compound having two – OH groups

$R-OH + Na \longrightarrow RONa + \frac{1}{2}H_2$

Total moles of H_2 liberated is $1\frac{1}{2}$ moles; occupies 33.6 lit.

48. Let V.P of pure A be P_A^0

Let V.P of pure B be P_B^0

When $X_A = 0.7$ and $X_B = 0.3$

$P_s = 350$

$\Rightarrow P_A^0 \times 0.7 + P_B^0 \times 0.3 = 350$

When $X_A = 0.2$ and $X_B = 0.8$

$P_s = 410$

$\Rightarrow P_A^0 \times 0.2 + P_B^0 \times 0.8 = 410$

Solving (i) and (ii)

$P_A^0 = 314$ mmHg

$P_B^0 = 434$ mmHg

$\therefore$ Answer = 314

49. Where dien is tridentate ligand

$H_2\ddot{N}-CH_2-CH_2-\ddot{N}H-CH_2-CH_2-\ddot{N}H_2$

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50. $2N_2O_5 \longrightarrow 4NO_2 + O_2$

Initial p_0 0 0

Final $p_0 - 2x$ $4x$ x

$$k = \frac{2.303}{t}\log\frac{p_0}{p_t}$$

$$3.38\times10^{-5} = \frac{2.303}{20\times60}\log\frac{100}{p_t}$$

$$0.0176 = \log\frac{100}{p_t}$$

$$p_t = 96 \text{ atm}$$

MATHEMATICS

51. $$T_r = \frac{3^{r-1}(4-2r)}{r(r+1)(r-1)}$$

$$T_r = \frac{3^{r-1}}{r(r-1)} - \frac{3^r}{r(r+1)}$$

$$\sum_{r=2}^{100} T_r = \frac{3^{2-1}}{2(2-1)} - \frac{3^2}{2.3} + \frac{3^2}{3.2} - \frac{3^2}{3.4} + ... + \frac{3^{99}}{100\times99} - \frac{3^{100}}{100\times101}$$

$$\sum_{r=2}^{100} T_r = \frac{3}{2} - \frac{3^{100}}{100\times101}$$

52. We know that $\sin t = \cos\left(\frac{\pi}{2} - t\right)$ and $\cos t = \sin\left(\frac{\pi}{2} - t\right)$

So, $\sin t + i\sin t = \cos\left(\frac{\pi}{2} - t\right) + i\sin\left(\frac{\pi}{2} - t\right)$

Applying De-Moivre theorem

Let $\theta = \frac{\pi}{2} - t$

$$(\sin + i\cos t)^n = \left(\cos\left(\frac{\pi}{2} - t\right) + i\sin\left(\frac{\pi}{2} - t\right)\right)^n$$

Using, D-Moivre theorem, this becomes $= \cos\left(n\left(\frac{\pi}{2} - t\right)\right) + i\sin n\left(\frac{\pi}{2} - t\right)$

This simplifies to

$$\cos\left(\frac{n\pi}{2} - nt\right) + i\sin\left(\frac{n\pi}{2} - nt\right)$$

$$\sin(nt) + i\cos(nt) = \cos\left(\frac{\pi}{2} - nt\right) + i\sin\left(\frac{\pi}{2} - nt\right)$$

For the identity to hold for all real t, we must have

$$\cos\left(\frac{n\pi}{2} - nt\right) + i\sin\left(\frac{n\pi}{2} - nt\right) = \cos\left(\frac{\pi}{2} - nt\right) + i\sin\left(\frac{\pi}{2} - nt\right)$$

This implies that $\frac{n\pi}{2} - nt = \frac{\pi}{2} - nt + 2k\pi$ for some integer k

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Simplifying, $\frac{n\pi}{2}=\frac{\pi}{2}+2k\pi$

Dividing by π, $\frac{n}{2}=\frac{1}{2}+2k$

Multiplying by 2, $n = 4k + 1$

We need $1\le 1+4k\le 1000$

Subtracting 1, $0 \le 4\,k \le 999$

Dividing by 4, $0\le k\le\frac{999}{4}$

Since k must be an integer $0 \le k \le 49$

This possible value for k are 0, 1, ……, 249

The number of such values is 249 – 0 + 1 = 250

53. $8\sin x\,\cos x(\cos 2x)=\sqrt{3}$

$$4\sin 2x\cos 2x=\sqrt{3}\Rightarrow \sin 4x=\frac{\sqrt{3}}{2}$$

$$4x=n\pi+(-1)^n\frac{\pi}{3}$$

$$x=\frac{n\pi}{4}+(-1)^n\frac{\pi}{12}$$

54. Make cases

Case 1: Non-zero solutions = $^7C_2.2\cdot 2\cdot 2=168$

Case 2: If exactly one of x, y, z is zero, number of solutions = $3\times 2\times 2\times {}^7C_1=84$

Case 3: If exactly two x, y, z is zero, number of solutions = 3 × 2 = 6

Total solutions = 258

55. Given $\left[(1-x)(1+x+x^2)\right]^n=\sum_{k=0}^{n}a_k x^k\frac{(1-x)^{3n}}{(1-x)^{2k}}$

$$\Rightarrow\left[\frac{1+x+x^2}{(1-x)^2}\right]^n=\sum_{k=0}^{n}a_k\frac{x^k}{(1-x)^{2k}}$$

$$=\left[\frac{(1-x)^2+3x}{(1-x)^2}\right]^n=\sum_{k=0}^{n}a_k\left(\frac{x}{(1-x)^2}\right)^k\Rightarrow\left[1+\frac{3x}{(1-x)^2}\right]^n=\sum_{k=0}^{n}a_k\left(\frac{x}{(1-x)^2}\right)^k$$

Put $\frac{x}{(1-x)^2}=t$

$$\Rightarrow(1+3t)^n=\sum_{k=0}^{n}a_k t^k$$

Now $3.a_{k-1}+a_k = 3.$ (coeff. of t^{k-1}) + coeff. of t^k

$= 3.3^{k-1}\,{}^nC_{k-1} + 3^k.{}^nC_k = 3^k\{{}^nC_{k-1} + {}^nC_k] = 3^k\,.\,{}^{n+1}C_k$

56. If the sum of powers of 3 and 7 is even, the divisor is of the form 4n + 1. If the sum of powers of 3 and 7 is odd the divisor is of the form 4n – 1. So, half of all the divisors will be of the form 4n +1.

57. Let $x + 5 = 14\cos\theta$ and $y - 12 = 14\sin\theta$, $\theta \in [0, 2\pi)$

$\therefore x^2 + y^2 = 365 + 28(12\sin\theta - 5\cos\theta) = 365 + 364\sin(\theta - \phi)$

Where $\tan\phi=\frac{5}{12}$

$\therefore$ Minimum value of $x^2 + y^2 = 1$

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58. $\begin{bmatrix}1 & \alpha\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & \beta\\ 0 & 1\end{bmatrix} = \begin{bmatrix}1 & \alpha+\beta\\ 0 & 1\end{bmatrix}$

$\Rightarrow 1 + 2 + 3 + \ldots\ldots + n = 378$

$\frac{n(n+1)}{2} = 378$

$n = 27$

59. $f(x) = \sin\{(2^x\})$ $\quad x \in [0, 2]$

$f(x) = \sin(2^x - [2^x])$

So, $[2^x] = 1, 2, 3, 4$

$f(0) = 0, f(0^+) = 0$

So, f(x) has 3 point of discontinuity

60. Hint: $f(x) - f(y) = \ln\frac{x}{y} + x - y$

$f(x) - f(y) \geq \ln x - \ln y + x - y$

$x = \alpha, y = \beta$

$\Rightarrow f(\alpha) - f(\beta) \geq \ln\alpha - \ln\beta + \alpha - \beta$ $\quad \ldots..(i)$

$x = \beta, y = \alpha$

$\Rightarrow f(\beta) - f(\alpha) \geq \ln\beta - \ln\alpha + \beta - \alpha$

$\Rightarrow f(\beta) - f(\alpha) \leq \ln\alpha - \ln\beta + \alpha - \beta$ $\quad \ldots..(ii)$

(i) and (ii)

$\Rightarrow f(\alpha) - f(\beta) = \ln\alpha - \ln\beta + \alpha - \beta$ $\quad \Rightarrow f(\alpha) - \ln x - \alpha = f(\beta) - \ln\beta - \beta$

$\Rightarrow f(x) - \ln x - x = k$(constant) $\quad \Rightarrow f(x) = \ln x + x + k$

$g(x) = f'(x) = 1 + \frac{1}{x}$

$\sum g\left(\frac{1}{n}\right) = \sum(1+n) = 100 + 5050 = 5150$

61. $f(x) = 1 + \frac{1}{\left(x + \frac{1}{x}\right)} + \left(x + \frac{1}{x}\right)$

Let $g(t) = t + \frac{1}{t} \Rightarrow g'(t) = 1 - \frac{1}{t^2} \geq 0$

$\Rightarrow t \leq -1$

$\because x > 0 \Rightarrow x + \frac{1}{x} \geq 2$

$\Rightarrow f(x) = 1 + \frac{1}{\left(x + \frac{1}{x}\right)} + \left(x + \frac{1}{x}\right)$ is increasing function when $x > 0$

$\Rightarrow$ f(x) is minimum when $x + \frac{1}{x} = 2$

$(f(x))_{min} = 3.5,$ when x = 1

62. $\because\ f(x+1) = \frac{1}{2} + \sqrt{f(x) - (f(x))^2}$

$\therefore\ f(x+2) = f(x+1+1) = \frac{1}{2} + \sqrt{f(x+1) - (f(x+1))^2} = \frac{1}{2} + \left|\frac{1}{2} - f(x)\right|$

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$\frac{1}{2}-\frac{1}{2}+f(x), \quad \left(\because \frac{1}{2} \le f(x) \le 1, \forall x\right) = f(x)$

63. We know that

$f(2n+3)+f(2n+7)=2$ ….. (1)

Replaced n by n + 1

$f(2n+5)+f(2n+9)=2$ ……(2)

Now replace n by n + 2

$f(2n+7)+f(2n+11)=2$ …….(3)

From (i) – (iii)

$f(2n+3)-f(2n+11)=0$

i.e. $f(2n+3)=f(2n+11)$

$\Rightarrow T=4$

64. $\left(A-\frac{I}{2}\right)\left(A-\frac{I}{2}\right)^T-I \Rightarrow \left(A-\frac{I}{2}\right)\left(A^T-\frac{I}{2}\right)=I$

$\Rightarrow AA^T-\frac{AI}{2}-\frac{A^TI}{2}+\frac{I}{4}=I$ …..(1)

$\left(A+\frac{I}{2}\right)\left(A+\frac{I}{2}\right)^T=I$

$\Rightarrow AA^T+\frac{AI}{2}+\frac{A^TI}{2}+\frac{I}{4}=I$ ……(2)

From (1) – (2) $AI + A^TI = 0 \Rightarrow A + A^T = 0$

$\Rightarrow A^T = -A \Rightarrow$ A is an skew symmetric

From (1) + (2), $AA^T=\frac{3I}{2} \Rightarrow |A| \neq 0$

$\Rightarrow$ A is skew symmetric matrix of even order

65. The number of ways of choosing 4 squares from 64 is $^{64}C_4$

The possible split ups as far as colour is concerned and the number of ways of selecting the square are listed below

White = 0, Black = 4

Number of ways = $^{32}C_0 \times {}^{32}C_4$

White = 1, Black = 3

Number of ways = $^{32}C_1 \times {}^{32}C_3$

White = 2, Black = 2

Number of ways = $^{32}C_2 \times {}^{32}C_2$

White = 3, Black = 1

Number of ways = $^{32}C_3 \times {}^{32}C_1$

White = 4, Black = 0

Number of ways = $^{32}C_4 \times {}^{32}C_0$

The number of ways in which 3 squares are of one colour and fourth is of opposite colour is

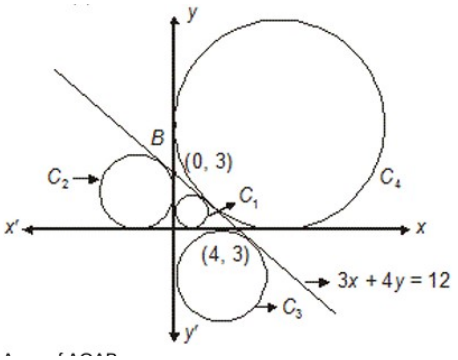
$P(A)=2\times {}^{32}C_1\times {}^{32}C_3 = (2\times 32)\frac{32\times 31\times 30}{2\times 3}$

The total number of ways of selecting 4 squares is $=\frac{64\times 63\times 62\times 61}{2\times 3\times 4}$

The required probability = $\frac{P(A)}{P(B)}=\frac{640}{1281}$

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66.



Area of $\Delta OAB = \frac{1}{2} \times 4 \times 3 = 6$ and $a = 3, b = 4, c = \sqrt{3^2 + 4^2} = 5$

$\therefore s = \frac{a+b+c}{2} = \frac{3+4+5}{2} = 6$

$\therefore$ Sum of radius of circles
= radius of circle C_1 + radius of circle C_2 + radius of circle C_3 + radius of circle C_4
$= r + r_1 + r_2 + r_3$

$$= \frac{\Delta}{s} + \frac{\Delta}{s-a} + \frac{\Delta}{s-b} + \frac{\Delta}{s-c} = \frac{6}{6} + \frac{6}{6-3} + \frac{6}{6-4} + \frac{6}{6-5} = 1 + 2 + 3 + 6 = 12$$

67. Let $P\left(t_1, \frac{1}{t_1}\right)$ and $Q\left(t_2, \frac{1}{t_2}\right)$

Let R(x, y) be a locus point.

Then $3x = 2t_1 + t_2$ .... (i)

$3y = \frac{2}{t_1} + \frac{1}{t_2} \Rightarrow 2t_2 + t_1 = 3yt_1t_2$ ....(ii)

Slope of $\overline{PQ} = 4 \Rightarrow \frac{\frac{1}{t_2} - \frac{1}{t_1}}{t_2 - t_1} = 4$

$\Rightarrow t_1t_2 = \frac{-1}{4}$ .....(iii)

From (iii)

$2t_2 + t_1 = \frac{-3y}{4} \Rightarrow 4t_1 + 8t_2 = -3y$

From (i)

$4t_1 + 2t_2 = 6x$

Solving, $t_2 = \frac{-2x - y}{2}$, $t_1 = \frac{8x + y}{4}$

Now, from (iii), $t_1t_2 = \frac{-1}{4}$

$\Rightarrow 16x^2 + 10xy + y^2 = 2$

68. Let $I_n = \int_0^{\pi} \frac{1 - \cos nx}{1 - \cos x} dx$

$I_{n+1} + I_{n-1} = \int_0^{\pi} \frac{2 - 2\cos nx \cos x}{1 - \cos x} dx$

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$$= 2\int_0^{\pi} \frac{1-\cos nx + \cos nx(1-\cos x)}{1-\cos x}dx \;= 2I_n + 2\int_0^{\pi} \cos n\,x dx$$

$$\Rightarrow I_n = \frac{I_{n-1} + I_{n+1}}{2}$$

So, $I_1, I_2, \ldots\ldots$ are in A.P.

$I_1 = \pi$, $I_2 = 2\pi$ and so on

Hence $I_8 = 8\pi$

69. $\vec{b}_1 = -4\hat{i} + 3\hat{j} + 2\hat{k},\ \vec{b}_2 = -4\hat{i} + \hat{j} + \hat{k},\ \vec{a}_1 = -3\hat{i} + 6\hat{j},\ \vec{a}_2 = -2\hat{i} + 7\hat{k}$

$$\left|\vec{b}_1 \times \vec{b}_2\right| = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -4 & 3 & 2 \\ -4 & 1 & 1 \end{vmatrix} = \left|\hat{i}(3-2) - \hat{j}(-4+8) + \hat{k}(-4+12)\right| \;= \sqrt{1+16+64} = 9$$

$$\vec{a}_2 - \vec{a}_1 = \hat{i} - 6\hat{j} + 7\hat{k}$$

$$\left[\vec{a}_2 - \vec{a}_1 \;\; \vec{b}_1 \;\; \vec{b}_2\right] = \left(\vec{a}_2 - \vec{a}_1\right) \cdot \left(\vec{b}_1 \times \vec{b}_2\right) \qquad = 1 + 24 + 56 = 81$$

$$\therefore \text{ Distance between skew lines } = \left|\frac{\left[\vec{a}_2 - \vec{a}_1 \;\; \vec{b}_1 \;\; \vec{b}_2\right]}{\left|\vec{b}_1 \times \vec{b}_2\right|}\right| \;= \left|\frac{81}{9}\right| \;= 9$$

70. $$I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{\sin x} \;= 2\int_0^{\pi} \frac{\sin nx}{\sin x}dx$$

$$I_{n+2} - I_n = 2\int_0^{\pi} \frac{\sin(n+2)x}{\sin x}dx - 2\int_0^{\pi} \frac{\sin\ nx}{\sin x}dx \qquad = 2\int_0^{\pi} \frac{\sin(n+2)x - \sin nx}{\sin x}dx$$

$$= 4\int_0^{\pi} \frac{\cos(nx+x)\cdot \sin x}{\sin x}dx \;= 4\int_0^{\pi} \cos(n+1)\ x dx \;= 4\left[\frac{\sin(n+1)x}{n+1}\right]_0^{\pi} = \frac{4}{n+1}\left[\sin(n+1)\pi\right] = 0$$

$$\therefore\ I_{n+2} = I_n$$

$$\Rightarrow I_n = \begin{cases} I_1 & \text{if } n \text{ is odd} \\ I_2 & \text{if } n \text{ is even} \end{cases}$$

$$\Rightarrow I_n = \begin{cases} 2\pi & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

$$\Rightarrow \sum_{i=10}^{20} I_i = (I_{10} + I_{12} + \ldots.. + I_{20}) + (I_{11} + I_{13} + \ldots\ldots I_{19}) = 0 + 10\pi = 10\pi$$

71. $$\sum_{r=0}^{10} \cos^3 \frac{\pi r}{3} = \frac{1}{4}\sum_{r=0}^{10}\left(\cos \pi r + 3\cos \frac{\pi r}{3}\right)$$

$$I_1 = \sum_{r=0}^{10} \cos \pi r \;\;\&\;\; I_2 = \sum_{r=0}^{10} \cos\left(\frac{\pi r}{3}\right)$$

$I_1 = 1$

$$I_2 = \sum_{r=0}^{10} \cos \frac{r\pi}{3} \;= \frac{\cos\left(\frac{10}{2}\cdot\frac{\pi}{3}\right)\cdot \sin\frac{11\pi}{6}}{\sin\frac{\pi}{6}} = -\frac{1}{2}$$

$$\therefore\ I = \frac{1}{4}\left(1 - \frac{3}{2}\right) \;= -\frac{1}{8}$$

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72. $\alpha^3 = 1 \Rightarrow 1, \omega, \omega^2$ are roots

73. Let $y = f(x) = \left(x^2 + \frac{250}{x}\right)$,

$\therefore \quad \frac{dy}{dx} = f'(x) = 2x - \frac{250}{x^2}$

Put $f'(x) = 0$

$\Rightarrow 2x^3 - 250 = 0 \quad \Rightarrow x^3 = 125$

$\Rightarrow x = 5$

Again, $\frac{d^2y}{dx^2} = f''(x) = 2 + \frac{500}{x^3}$

Now $f''(5) = 2 + \frac{500}{125} > 0$

Hence at x = 5. The function will be minimum

Minimum value f(5) = 25 = 50 = 75

74. Hint: $\int_{-\pi/2}^{\pi/2} \sin^2 x \cdot \cos^2 x (\sin x + \cos x)\,dx$

$= \int_{-\pi/2}^{\pi/2} \sin^3 x \cdot \cos^2 x\,dx + \int_{-\pi/2}^{\pi/2} \sin^2 x \cdot \cos^3 x\,dx$

Solution:

$I = 0 + 2\int_0^{\pi/2} \sin^2 x \cdot \cos^3 x\,dx$

$\because = 2\frac{1\times 2}{5\times 3\times 1} = \frac{4}{15} = \frac{m}{n}$

$\Rightarrow m + n = 19$

75. Use sum of $\sum n$, $\sum n^2$ and $\sum n^3$

$$\lim_{n\to\infty} \frac{n \cdot \frac{\overbrace{n(n+1)}}{2} \cdot \frac{n(n+1)(2n+1)}{6} \cdot \frac{\overbrace{n^2(n+1)^2}}{4}}{n^8\left[\frac{1^4}{n^4} + \frac{2^4}{n^4} + \frac{3^4}{n^4} + \ldots + \frac{n^4}{n^4}\right]^2}$$

$$\lim_{n\to\infty} \frac{\frac{1}{2}\left(1+\frac{1}{n}\right)\frac{1}{6}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)\frac{1}{4}\left(1+\frac{1}{n}\right)^2}{\frac{1}{n^2}\left[\frac{1^4}{n^4} + \frac{2^4}{n^4} + \frac{3^4}{n^4} + \ldots + \frac{n^4}{n^4}\right]^2} \Rightarrow \lim_{n\to\infty} \frac{\frac{1}{2}\left(1+\frac{1}{n}\right)\frac{1}{6}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)\cdot\frac{1}{4}\left(1+\frac{1}{n}\right)^2}{\left(\frac{1}{n}\sum_{r=1}^{n}\left(\frac{r}{n}\right)^4\right)^2}$$

$$\lim_{n\to\infty} \frac{\frac{1}{2}\left(1+\frac{1}{n}\right)\frac{1}{6}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)\cdot\frac{1}{4}\left(1+\frac{1}{n}\right)^2}{\left(\int_0^1 x^4\,dx\right)^2} = \frac{1}{2}\times\frac{1}{6}\times 2\times\frac{1}{4}\times 25 = \frac{25}{24}$$

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